Taiwan Experience Education Program (TEEP): Research, Technology, Culture, and Life in Taiwan

Shiv Kumar Gupta

1. Introduction

This report presents my experience as a participant in the Taiwan Experience Education Program (TEEP), focusing on my research internship as well as my experiences of living, learning, and exploring Taiwan. The main focus of my internship was the development and evaluation of a Robot Nurse Helper using the Kebbi social robot, combining artificial intelligence, computer vision, robotics, and healthcare. During the internship, I worked on features including English and Chinese interaction, cataract-related information, and object detection, followed by testing in a hospital environment. In addition to the research work, I had opportunities to experience Taiwan's academic and technological environment through activities such as visiting Taipei 101 and attending the AIWAVE exhibition. I also explored various cultural, historical, and natural destinations across Taiwan. These experiences helped me develop my technical and research skills while also giving me a better understanding of Taiwanese culture, daily life, and the importance of international exchange.

2. Research Project: Robot Nurse Helper Using Kebbi

The primary project of my internship was the development and evaluation of a Robot Nurse Helper using the Kebbi social robot. The project explored how an interactive robot could be used in a healthcare environment to support patients and provide useful information and assistance. It brought together several areas of technology, including artificial intelligence, computer vision, robotics, and human-robot interaction.

During the project, I worked on different features of the Kebbi system. One important aspect was creating interaction in both English and Chinese, allowing the robot to communicate with users in different languages. We also developed functions related to cataract surgery, with the aim of providing patients with relevant information and assistance. Another major component was object detection, which introduced computer vision capabilities and allowed the robot to recognize objects in its surroundings.

Working on this project gave me practical experience in integrating multiple technologies into a physical robotic platform. It also helped me understand that healthcare technology needs to be designed not only from a technical perspective but also according to the needs, comfort, and understanding of its users.

3. Hospital Testing and Evaluation

A particularly important part of the internship was the opportunity to test the Kebbi Robot Nurse Helper in a real hospital environment. This was an important step because the system had previously been developed and tested in a more controlled setting. Using the robot in a hospital allowed us to observe how it performed in a real healthcare environment and how people responded to its interaction.

The testing provided encouraging feedback and showed that the concept had potential for practical healthcare applications. At the same time, the experience helped us identify areas that could be improved. Some aspects of the interaction and system behavior required further refinement when considered from the perspective of actual users and the hospital environment.

This experience taught me the importance of real-world testing and user feedback in technology development. A system may work successfully during development, but its effectiveness can only be properly understood when it is used in the environment for which it was designed. The hospital evaluation therefore became an important part of the development cycle, helping us understand what worked well and what could be improved.

4. Research and Collaboration Experience

The internship provided me with the opportunity to work in an interdisciplinary and international research environment. The Kebbi project required knowledge from different areas, and working with professors and students helped me understand how different perspectives can contribute to the development of a single project.

Throughout the internship, I gained experience in discussing technical problems, receiving feedback, testing different approaches, and working toward a common research goal. I also learned that research involves continuous adjustment rather than simply following an original plan. Technical problems, user feedback, and observations from testing can all lead to changes in the system.

This experience improved my ability to communicate technical ideas, collaborate with others, solve problems, and adapt to different working styles. It also gave me a better understanding of how artificial intelligence can be connected with real-world applications, particularly in healthcare.

5. Academic and Technology Experiences

TEEP also gave me opportunities to learn outside the laboratory. One of the most interesting experiences was attending the AIWAVE exhibition, where I was able to see a variety of emerging technologies and applications related to artificial intelligence. As someone working on an AI and robotics project, the exhibition was especially useful because it allowed me to observe technologies beyond my own research and understand how AI is being applied in different fields.

The exhibition encouraged me to think about the future possibilities of artificial intelligence, computer vision, robotics, and intelligent systems. It also helped me connect the concepts I had learned through my academic studies with technologies being developed for practical applications.

Another memorable activity was a visit to Taipei 101, organized through our internship experience with our professor and the host institution. The visit provided an opportunity to experience one of Taiwan's most well-known landmarks and also allowed us to spend time together outside the research environment. These activities made the internship more balanced by combining academic learning with cultural and social experiences.

6. Living in Taiwan

Living in Taiwan was an important part of my overall experience. I found the country peaceful, organized, and comfortable to live in, and I enjoyed discovering something new almost every day. Visiting places such as Longshan Temple gave me an opportunity to learn about Taiwanese traditions, beliefs, architecture, and local culture. I also visited Chiang Kai-shek Memorial Hall, which helped me learn more about Taiwan's history and its important cultural and historical landmarks.

One of the things that impressed me most was the organization of daily life. The MRT and bus systems were very convenient, and transportation made it easy to travel between different parts of the city and beyond. The clear schedules and reliable timing made planning trips much easier. I also noticed that people generally followed well-organized queues at MRT stations, attractions, restaurants, and other public places. This orderly behavior and respect for shared spaces contributed to my impression of Taiwan as a peaceful and well-managed country.

Living in a different country also helped me become more independent. Everyday activities such as using public transportation, ordering food, visiting unfamiliar places, communicating with local people, and planning trips became opportunities to learn and adapt.

7. Travel and Exploration in Taiwan

During my free time, I explored many different parts of Taiwan. These trips allowed me to experience the country's natural beauty, historical places, traditional culture, and modern urban environment. I visited Qingtiangang Grassland, Jiufen Old Street, Shifen Waterfall, Golden Waterfall, Tamsui, Beitou, Chiang Kai-shek Memorial Hall, Longshan Temple, and several other destinations.

The natural destinations provided a refreshing contrast to my research work. Qingtiangang Grassland offered beautiful mountain and grassland scenery, while Shifen Waterfall and Golden Waterfall allowed me to experience Taiwan's distinctive natural landscapes. Jiufen Old Street provided a different experience through its traditional streets, local food, shops, and historic atmosphere.

I also enjoyed visiting Tamsui and Beitou, which showed different aspects of life around Taipei. Along with these destinations, exploring temples, historical landmarks, local streets, and other attractions helped me understand that Taiwan has a strong combination of modern development, traditional culture, and natural beauty.

Traveling was also an important part of becoming more independent. I learned how to plan routes, use public transportation, manage time, and adapt when plans changed. The convenience of Taiwan's transportation system made it possible to explore many places without needing a private vehicle.

8. Taiwanese Culture and Daily Life

One of the most interesting aspects of my stay was observing Taiwanese culture through everyday experiences. Rather than learning about culture only through formal activities, I experienced it through transportation, food, temples, public places, traditional streets, and interactions with people.

The organized queues I observed in many places were particularly noticeable. People generally waited patiently for buses, trains, attractions, and services, creating an orderly environment even in busy locations. I also appreciated the cleanliness and organization of public spaces.

Trying Taiwanese food and visiting local markets and streets gave me another way to experience the culture. These simple daily experiences helped me understand Taiwan more naturally and made my stay feel more connected to the local environment.

Overall, I enjoyed the peaceful atmosphere of Taiwan and appreciated the opportunity to experience a lifestyle different from my own. The combination of modern infrastructure, traditional culture, natural surroundings, and friendly everyday experiences made living in Taiwan memorable.

9. Challenges and Learning

My internship also came with challenges. Adapting to a new country, unfamiliar surroundings, different communication styles, and a new academic environment required time and flexibility. Working on a robotic system also involved technical challenges, particularly when integrating different features and making the system suitable for real-world use.

The hospital testing experience taught me that technical development does not end when a system starts functioning. Real users can provide perspectives that are difficult to predict during development, making evaluation and improvement essential.

I also learned to become more comfortable with unfamiliar situations. Traveling independently, navigating public transportation, communicating across language differences, and adjusting to a different environment improved my confidence and independence.

These experiences helped me develop skills beyond programming and technical work, including communication, teamwork, problem-solving, adaptability, and time management.

10. Conclusion

Overall, my TEEP experience was a valuable combination of research, technology, cultural exchange, and personal growth. It allowed me to work on a meaningful healthcare robotics project, experience research in an international environment, and learn how technology can be evaluated and improved through real-world interaction. At the same time, living in Taiwan gave me the chance to understand a new culture, explore different places, and become more independent in my everyday life. The peaceful environment, convenient transportation, organized public spaces, and many opportunities to learn something new each day made my stay especially memorable.

I am thankful to my professor, the host institution, and the TEEP program for providing this opportunity and supporting my academic and personal development. The knowledge, practical skills, cultural understanding, and experiences I gained during my time in Taiwan will remain valuable as I continue my studies and future career.